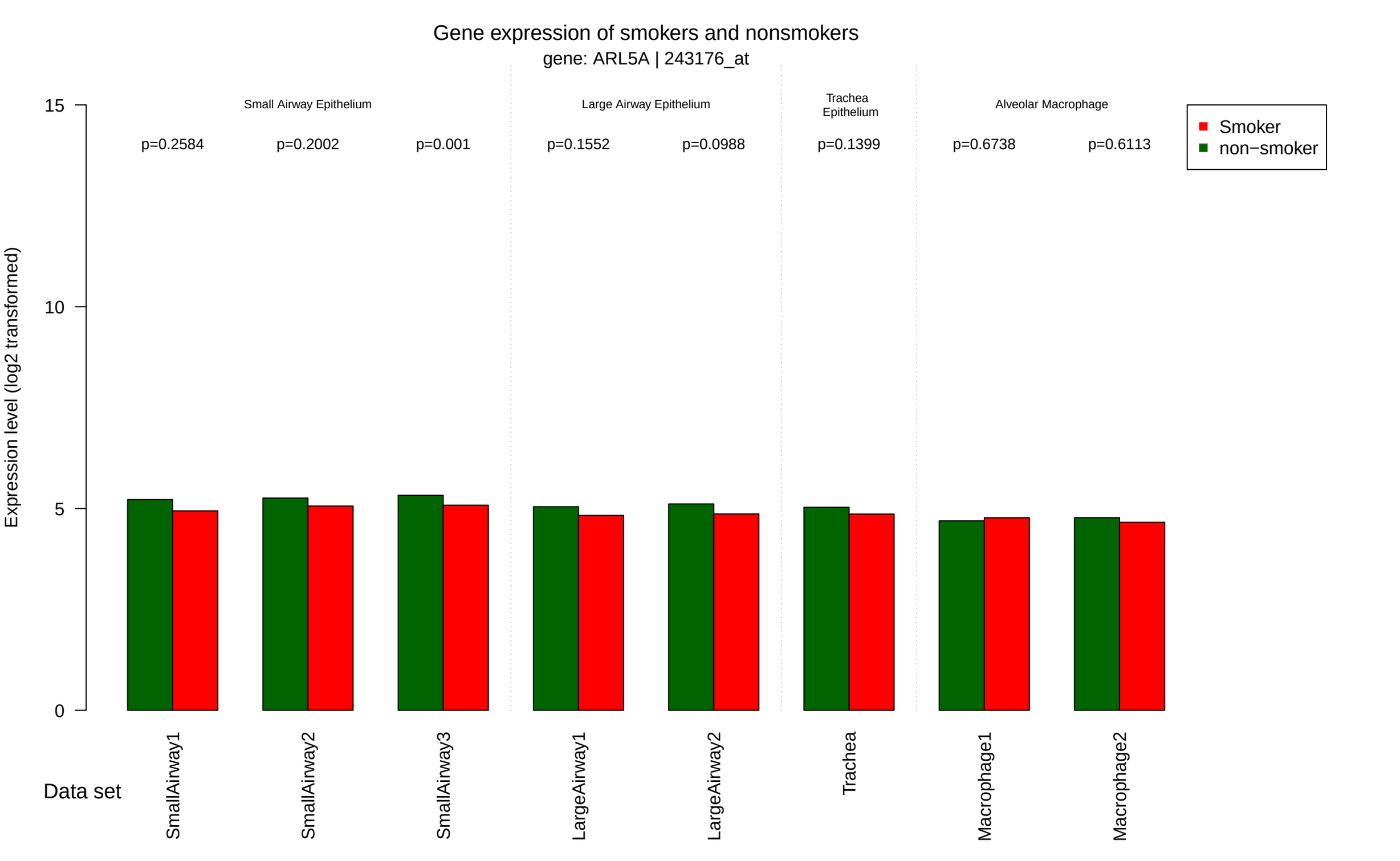
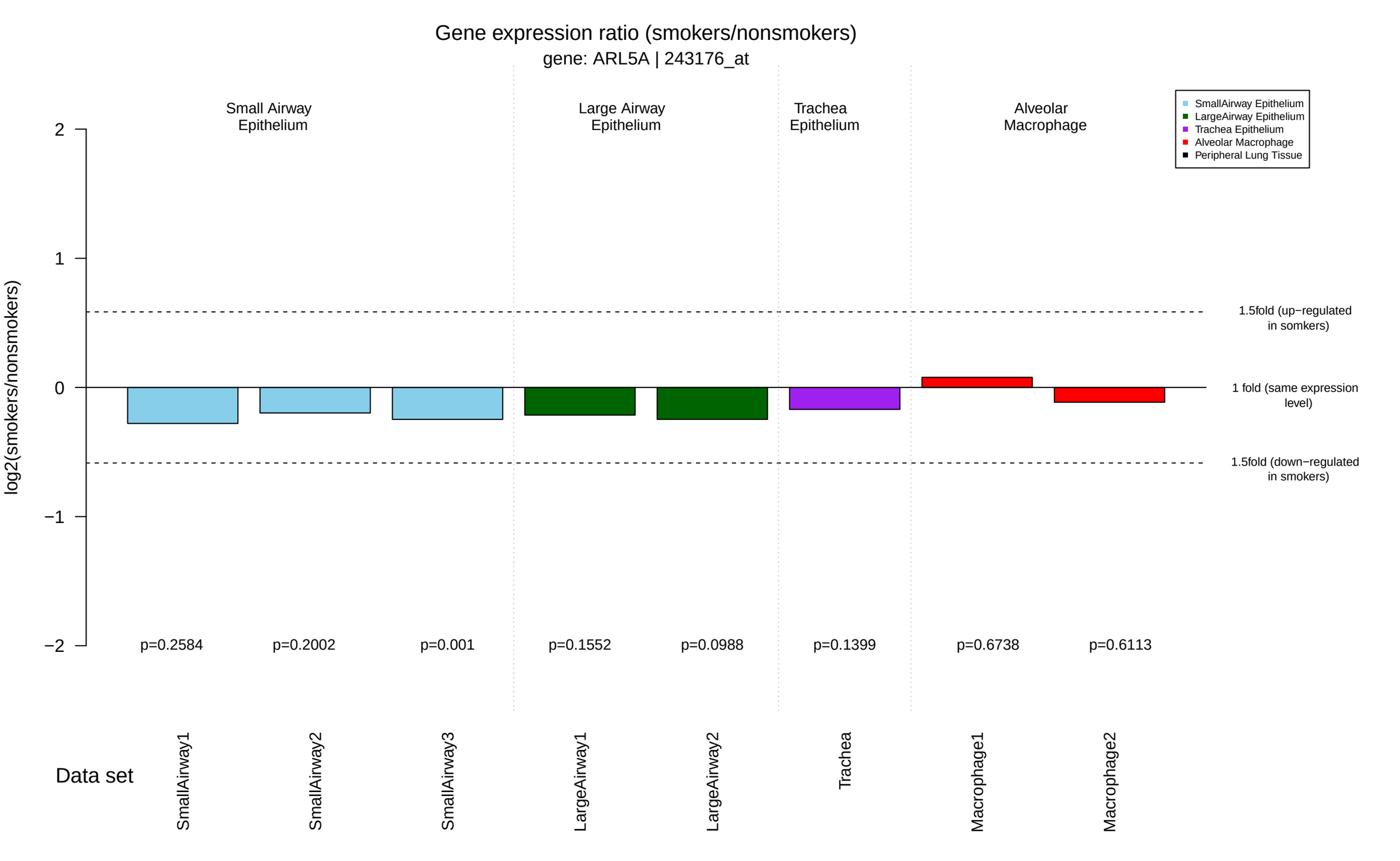
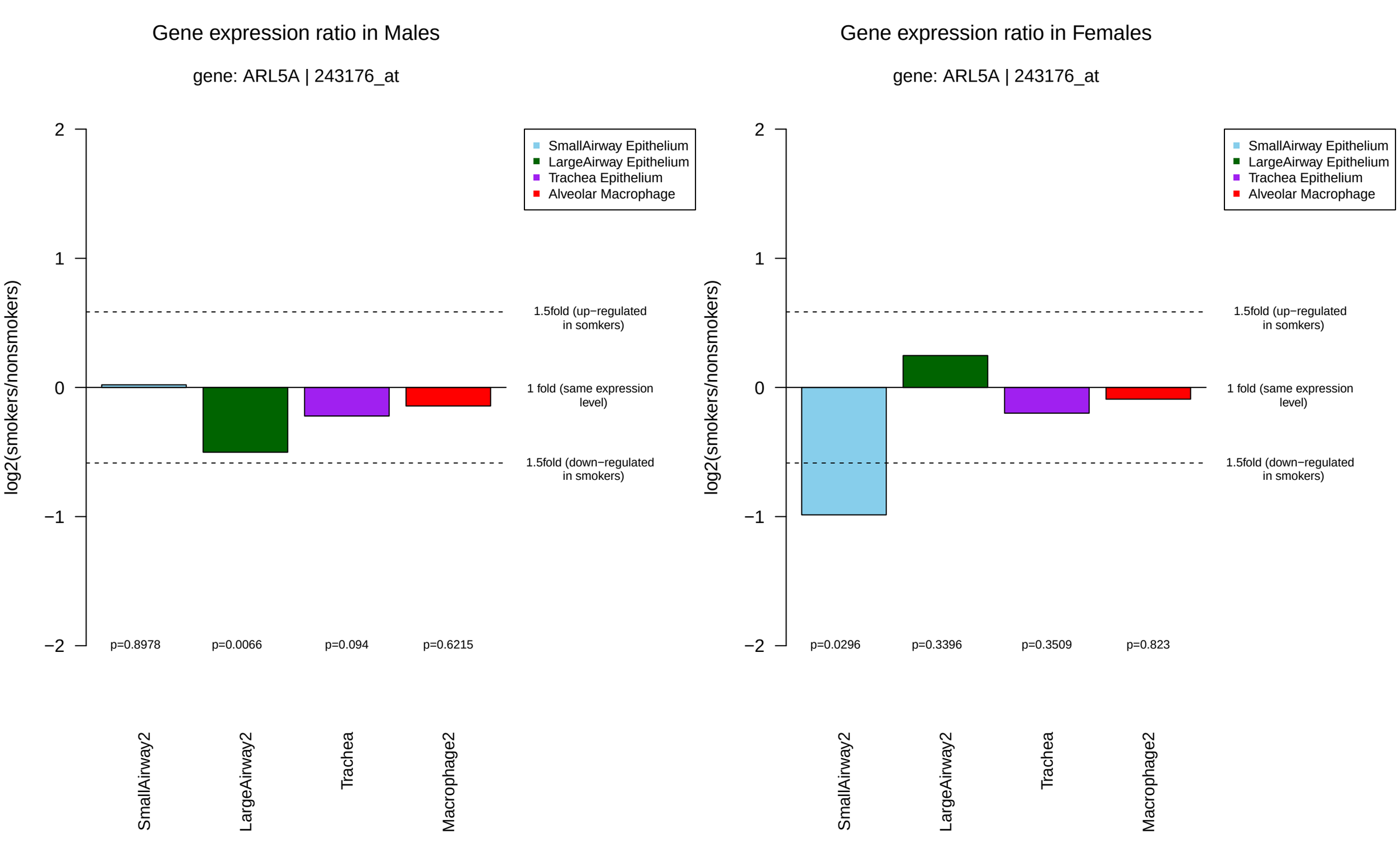
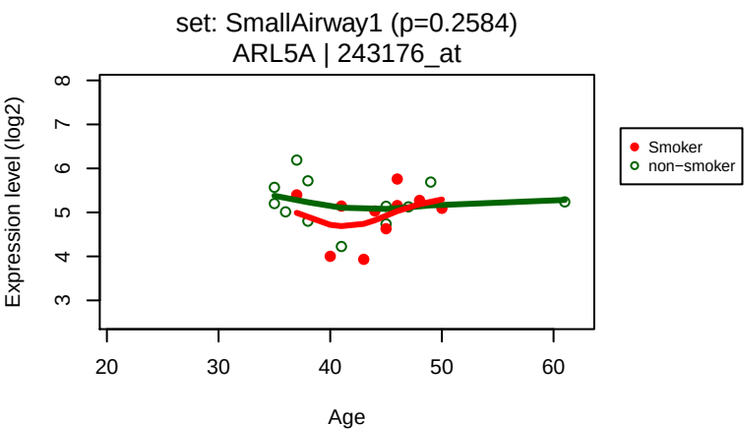
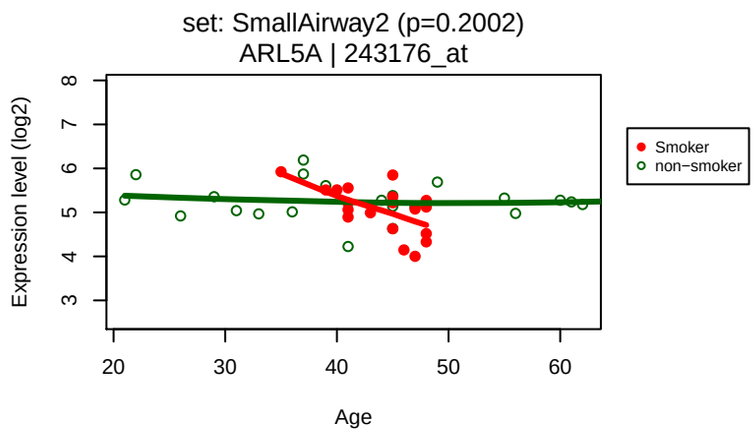
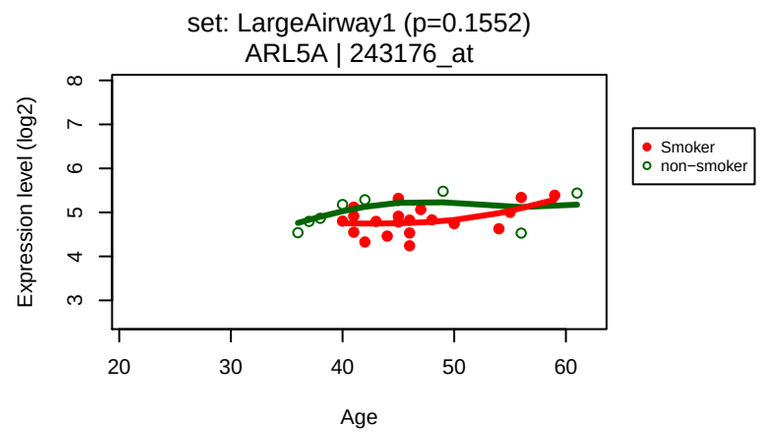

Gene expression of smokers and nonsmokers
gene: ARL5A | 243176_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2584
p=0.2002
p=0.001
p=0.1552
p=0.0988
p=0.1399
p=0.6738
p=0.6113
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

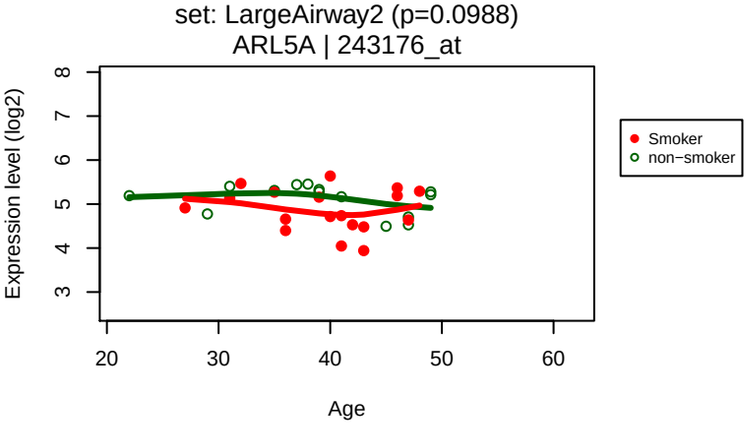

Gene expression ratio (smokers/nonsmokers)
gene: ARL5A | 243176_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2584
p=0.2002
p=0.001
p=0.1552
p=0.0988
p=0.1399
p=0.6738
p=0.6113
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

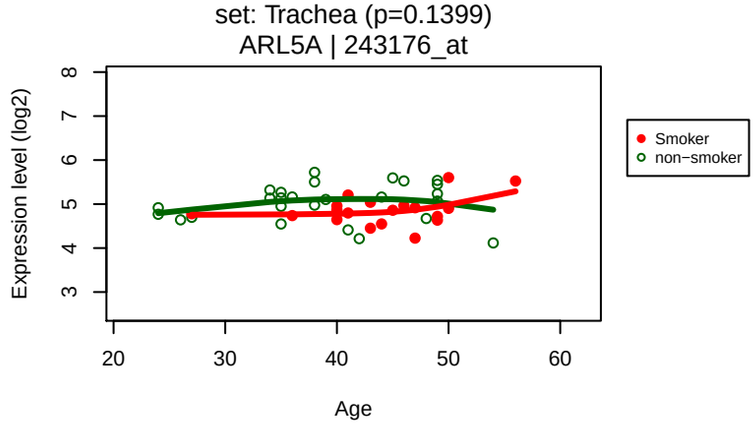

Gene expression ratio in Males
gene: ARL5A | 243176_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8978
p=0.0066
p=0.094
p=0.6215
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL5A | 243176_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0296
p=0.3396
p=0.3509
p=0.823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

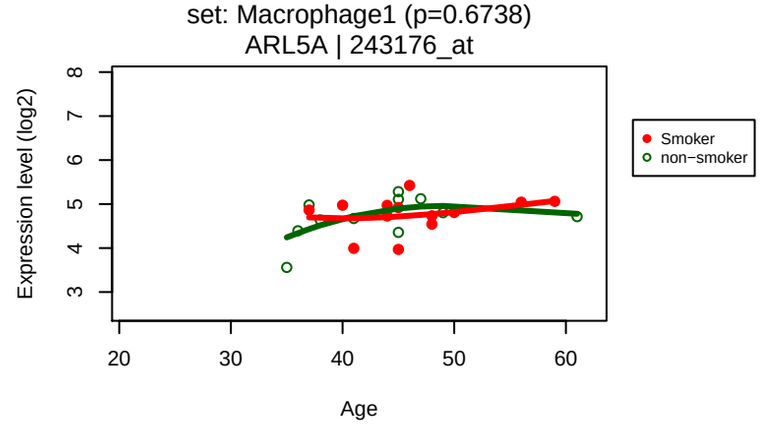

set: SmallAirway1 (p=0.2584)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker

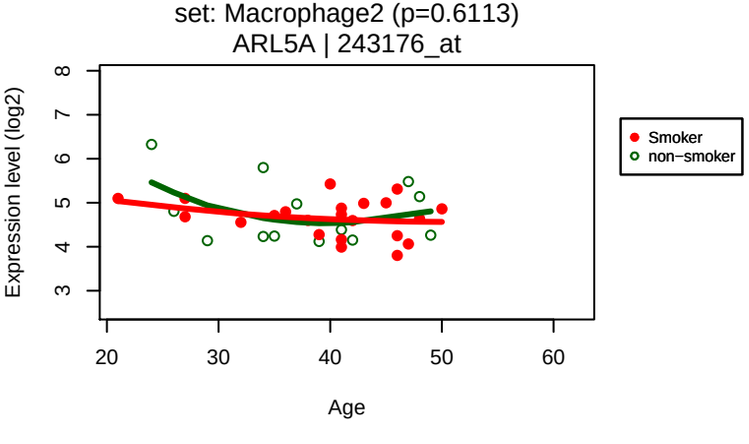

set: SmallAirway2 (p=0.2002)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker

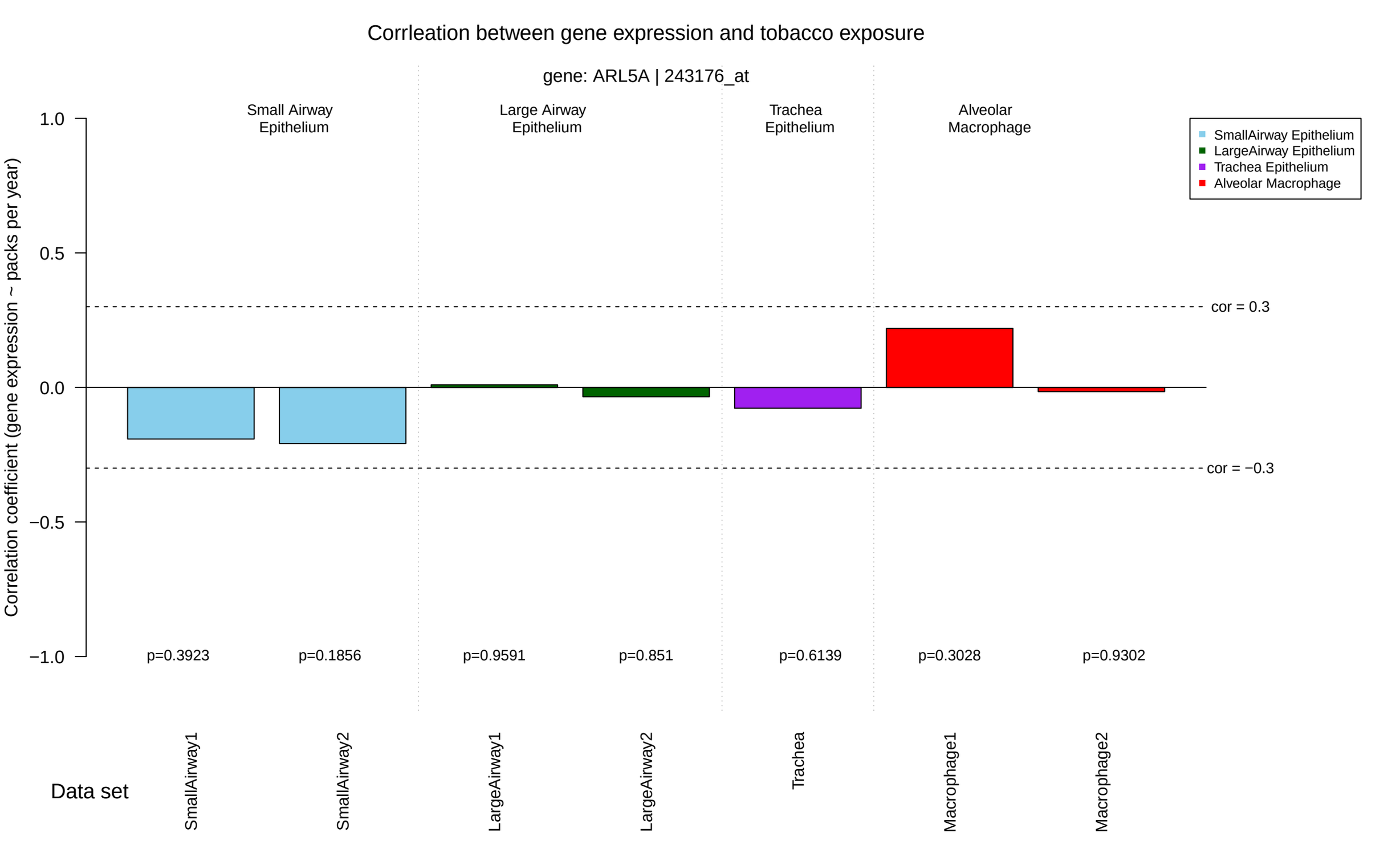

set: LargeAirway1 (p=0.1552)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0988)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1399)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6738)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6113)
ARL5A | 243176_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL5A | 243176_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3923
p=0.1856
p=0.9591
p=0.851
p=0.6139
p=0.3028
p=0.9302
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2